

Abolfazl Sajadi

Digital Hardware Engineer · Hardware Security

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Profile

Digital hardware engineer completing a Ph.D. at Leiden University, specializing in **RTL/SoC design** (SystemVerilog, VHDL), FPGA prototyping, ASIC implementation, and hardware security. Built the peripheral, control, and verification infrastructure of a dual-core RISC-V cryptographic SoC and carried the platform through **GlobalFoundries 22FDX** MPW tape-out, package/PCB design, and post-silicon bring-up to a fully tested, documented, working chip with its full software stack (Python/C libraries, GUI, CLI).

Experience

Aug 2022 –
Dec 2026

● ONGOING

Doctoral Researcher - RTL/SoC, ASIC & Hardware Security Leiden University, LIACS · Leiden, NL

End-to-end SoC development. Drove PROACT from top-level RTL assembly and custom RISC-V peripherals through verification, FPGA prototyping, GF 22FDX tape-out, packaging, PCB design, and post-silicon bring-up of the fabricated chip.

- **RTL and RISC-V control:** Integrated two Ibex RISC-V cores with third-party AES, Ascon, and Xoodoo co-processor RTL; designed the surrounding memory-mapped I/O, control/status registers, command interfaces, dedicated SPI bridge and UART (runtime-configurable baud) peripherals, CDC synchronizers, and reset-control architecture.
- **Embedded control software:** Defined the memory map and register protocol; built C/C++ firmware, Python host tools, and a GUI for program upload, configuration, register access, and cryptographic control.
- **Verification automation:** Built RTL regressions that generated and compiled randomized C tests, produced and loaded VMEM images, and analyzed logs to verify peripherals, co-processors, and system-level behavior before hardware prototyping.
- **FPGA prototyping (Vivado):** Deployed the complete platform on Xilinx Artix-7 and Zynq-7000 devices; validated the memory map, peripherals, co-processor control paths, and SPI/UART communication in hardware before ASIC implementation.
- **PPA and side-channel analysis:** Benchmarked countermeasures on power, performance, area, security, and cost; automated CPA, TVLA, SNR/NICV, and pre-silicon leakage-analysis flows.
- **ASIC implementation:** Synthesized the SoC for **GlobalFoundries 22FDX** in Cadence Genus, ran SDF-annotated gate-level simulation (Xcelium), and performed initial floorplanning, **place and route**, STA, DRC, and LVS; built a four-bank SRAM subsystem with Ibex access logic.
- **Tape-out and packaging:** Collaborated with imec on layout optimization, sign-off, and GDSII preparation for the $2.009 \times 2.055 \text{ mm}^2$ MPW die; prepared the chip/package pin mapping and 28-pin DIL bonding plan.
- **PCB design:** Created a **four-layer ChipWhisperer CW308-compatible Altium board** with power, clock/reset, UART/SPI, and measurement interfaces for **post-silicon bring-up** and testing.
- **Post-silicon validation:** Brought up, tested, and documented the chip: cores, co-processors, and peripherals working in silicon; now analyzing power traces for side-channel evaluation.

PROACT: [project website](#) · [Leiden University](#) · [code & documentation release forthcoming](#)

2018 – 2021

Research Assistant - Embedded Systems (full-time) University of Tehran, DVD-ES Lab · Tehran, IR

- Designed and evaluated **ML-resistant PUF authentication** protocols (DC-PUF, SQ-PUF) on FPGA; implemented CNN/SVM modeling attacks in Python (published in *J. Network & Computer Applications*).

Selected Projects

Python

GNN-based pre-silicon power-trace generation

Under submission

- Spatiotemporal **Graph Neural Network** (GCN + dilated TCN) predicting gate-level power traces **up to 9× faster than Cadence Joules**, accurately enough for full **AES-128 key recovery** by CPA.

Open-source

ASSESS - pre-silicon side-channel leakage analysis

ACM CF'26

- A cycle-accurate methodology that localizes **gate-level side-channel leakage** from toggle activity, fast enough for routine checks during iterative **ASIC design**; benchmarked against RTL-PAT, PATCH and ACA on RISC-V/AES designs.

Vivado HLS	Real-time object detection & tracking on Zynq-7000 – HW/SW co-design on an ARM+FPGA SoC: HDMI video in, filtering/tracking synthesized via High-Level Synthesis , output over VGA/UART.	B.Sc. thesis
Cyclone II	CNN accelerator on FPGA – RTL implementation of a tiny convolutional-network accelerator (MNIST), validated against a Keras reference model and brought up on an Altera Cyclone II board.	RTL / Quartus
ESP32 · AVR	Home meter monitoring device – Embedded design on ESP32 and AVR (ATmega8) microcontrollers, with a custom PCB designed in Altium Designer.	Altium · 2020

Education

Aug 2022 – Dec 2026 ● ONGOING	Ph.D. in Computer Science Hardware security & digital IC design (PROACT). Dissertation: pre/post-silicon side-channel leakage evaluation in lightweight cryptographic circuits.	Leiden University (LIACS) · Leiden, NL
2018 – 2021	M.Sc. in Electrical Engineering - Digital Electronic Systems Ranked 1st in class. Thesis: ML-based modeling attacks on Physical Unclonable Functions (PUFs).	University of Tehran · GPA 18.28/20
2014 – 2017	B.Sc. in Electrical Engineering - Electronic Technology GPA 16.87/20 · Top of class. Thesis: real-time object detection & tracking on a Zynq-7000 SoC.	Mohajer Technical University

Selected Publications

- **A. Sajadi**, N. Zidarič, T. Stefanov, N. Mentens. *Systematic Comparison and Improvement of Pre-silicon Leakage Analysis Tools*. ACM Computing Frontiers (CF Companion '26), 2026. [\[doi\]](#)
- **A. Sajadi**, N. Zidarič, T. Stefanov, N. Mentens. *A Systematic Comparison of Side-channel Countermeasures for RISC-V-based SoCs*. IEEE NorCAS, 2024. [\[doi\]](#)
- **A. Sajadi**, A. Shabani, B. Alizadeh. *DC-PUF: ML-Resistant PUF-Based Authentication for Resource-Constrained IoT*. J. Network & Computer Applications, vol. 217, 2023. [\[doi\]](#)
- A. Adhikary et al. (incl. **A. Sajadi**). *PROACT: Physical Attack Resistance of Cryptographic Algorithms and Circuits with Reduced Time to Market*. Applied Reconfigurable Computing (ARC), LNCS, Springer, 2024. [\[doi\]](#)
- J. Anders et al. (incl. **A. Sajadi**). *A Survey of Developments in Testability, Safety and Security of RISC-V Processors*. IEEE ETS, 2023. [\[doi\]](#)

Technical Skills

HDL & HLS	VHDL, Verilog, SystemVerilog, High-Level Synthesis (HLS)
Programming	Python, C/C++, Tcl (EDA scripting)
ASIC Front-End	RTL design, RISC-V (Ibex) integration, logic synthesis, SDC constraints, STA, gate-level simulation
Back-End	Place-and-route, sign-off, GDSII, bonding diagram (GF 22FDX)
Verification	RTL testbenches, functional simulation, FPGA validation, hardware debugging
FPGA/Embedded	Xilinx FPGAs, Vivado & Vivado HLS, Catapult HLS, SPI/UART, ESP32, AVR
HW Security	CPA, DPA, TVLA, SNR/NICV, ChipWhisperer, pre-silicon leakage analysis, PUFs
Design Methods	HW/SW co-design, continuous integration (CI), test & testability design (DfT), fault-tolerant system design
ML & EDA	Cadence (Genus, Joules, Xcelium), Altium, MATLAB, PyTorch, Keras

Teaching, Supervision & Activities

- **Teaching Assistant**, *System & Software Security*, Leiden University (2022–present); **co-supervised 6+ B.Sc. theses** on AES/RISC-V side-channel analysis and chip-prototype leakage assessment.
- **Cadence-certified** in RTL-to-GDSII Flow, Genus Physical Synthesis, and Tcl for EDA (2024–25); **Reviewer**, *IEEE Trans. Computer-Aided Design (TCAD)*.

Honors & Languages

- **Honors**: Ranked 102nd of 30,000+ in the national M.Sc. entrance exam; 1st place, provincial WorldSkills electronics competition (2013).
- **Languages**: Persian (native) · English (professional working) · Dutch (A2).